Self-Regulated Learning Interventions: Seeding Effects and Behavioral Side-Effects

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Abstract

This study examines how targeted interventions shape self-regulated learning (SRL) behaviors in asynchronous online courses through both immediate and longitudinal mechanisms. Using a randomized controlled trial design with 271 business students, we implemented two complementary interventions: an initial survey-based intervention providing personalized SRL guidance to students with low self-assessment scores, and a late-of-schedule (LoS) intervention alerting students who fell behind their self-determined deadlines. We analyzed both self-reported perceptions and behavioral learning analytics to map the dynamics of the learning process. Results revealed that the initial intervention created a "seeding effect"—while showing limited immediate behavioral changes, it altered students' developmental trajectories such that group differences emerged weeks later. The LoS intervention triggered a metacognitive awakening: students in the experimental group significantly revised their deadline perceptions ($p < .001$) and exhibited more feedback-seeking behaviors and increases in self-assessment activities compared to controls. Process analysis revealed that students in the experimental group developed more complex, self-regulatory learning patterns with reflective states serving as central hubs connecting different learning activities. These findings demonstrate that SRL interventions work not only through immediate correction but by fundamentally altering how students' learning processes evolve over time, while also producing unexpected redistributions of learning effort.

Keywords: self-regulated learning, intervention, transition network analysis, survey, learning analytics

1 Introduction

Self-regulated learning (SRL) has emerged as a critical determinant of success in online education, where students must manage their own learning processes with minimal external structure (Dent & Koenka, 2016). However, substantial individual differences exist in students' capacity for self-regulation, and many students struggle to effectively monitor their progress, seek feedback, and adjust their learning strategies without support (Hung et al., 2024; Rasheed et al., 2020). This challenge is particularly acute in asynchronous online courses, where the absence of scheduled class meetings removes natural regulatory cues and increases demands on students' metacognitive skills.

Recent research has demonstrated that targeted interventions can enhance self-regulated learning behaviors, but questions remain about the mechanisms through which such interventions operate (Braad et al., 2026) and whether their effects persist over time (Kizilcec et al., 2020). Most intervention studies focus on immediate outcomes, examining behavioral changes shortly after the intervention is delivered. However, this approach may miss important longitudinal effects—interventions might fundamentally alter how students' learning processes develop over the course duration (Mevarech & Amrany, 2008) rather than simply producing immediate behavioral shifts.

Additionally, the timing and nature of SRL interventions matter. General interventions delivered to all students may be inefficient, while targeted interventions addressing specific regulatory breakdowns (such as falling behind schedule) may be more effective (Dignath & Büttner, 2018). Yet we know little about how different types of interventions—broad skill-building versus targeted corrective—interact when deployed in sequence, or whether early interventions create conditions that enhance or diminish the effectiveness of later supports (Jansen et al., 2019).

This study addresses these gaps by examining a dual-intervention approach in an asynchronous online project management course. We implemented two complementary interventions: (1) an initial survey-based intervention that provided personalized SRL guidance to students with low self-assessment scores at course start, and (2) a late-of-schedule (LoS) intervention that alerted students who fell behind their self-determined deadlines during the course. Importantly, students in this course set their own submission schedules as their first assignment, making the concept of "lateness" personally meaningful rather than imposed externally (Ryan & Deci, 2020).

Our study makes three key contributions. First, we demonstrate that interventions can work through a "longitudinal seeding" mechanism—the initial intervention showed limited immediate behavioral effects but created differences that emerged weeks later, suggesting it altered students' developmental trajectories. Second, we reveal that targeted LoS notifications trigger what we term a "metacognitive awakening," prompting students to fundamentally reassess their learning processes rather than simply work faster. Third, through detailed sequence and network analysis of learning behaviors, we document both the benefits and trade-offs of intervention: while students in the experimental group developed more sophisticated self-regulatory patterns, they also shifted effort away from certain content engagement activities.

Research Questions:

1. How does an initial survey-based SRL intervention alter students' immediate learning processes and perceptions?
2. What are the longitudinal effects of the early interventions later during the course?
3. What kind of effects do LoS intervention targeting students who fall behind their self-determined schedules produce?
4. What behavioral patterns characterize students who receive intervention support, and what trade-offs emerge in their learning activity distributions?

2 Background

2.1 Self-Regulated Learning in Online Education

Self-regulated learning refers to the processes by which learners actively monitor, control, and regulate their cognition, motivation, and behavior to achieve learning goals. Zimmerman's (Zimmerman, 2000) cyclical model describes SRL as involving three phases: forethought (goal-setting and strategic planning), performance (self-monitoring and control), and self-reflection (self-evaluation and adaptation). In traditional classroom settings, instructors and course structures provide external regulation—scheduled classes, visible peer progress, and instructor monitoring create a regulatory scaffolding that supports students' self-regulation.

Online asynchronous courses fundamentally alter this regulatory landscape. Students must generate their own regulatory structures with minimal external cues, making strong SRL skills essential for success. Research consistently shows that students with stronger SRL skills achieve better outcomes in online courses, while those with weaker skills often struggle or withdraw (Broadbent & Poon, 2015). However, SRL skills are not static traits but rather dynamic capabilities that can be developed through instruction and support (Bellhäuser et al., 2016).

2.2 The Challenge of Individual Differences

A persistent challenge in online education is the substantial heterogeneity in students' SRL capacities. Some students arrive with well-developed metacognitive and strategic skills, while others lack awareness of effective learning strategies or their own regulatory needs. Recent longitudinal research by Kryshko et al. (2026) underscores this complexity, revealing that motivational regulation is a dynamic process where trajectories of strategy use differ significantly between students. Their findings highlight that individual background characteristics—particularly conscientiousness and prior academic achievement—strongly predict not only the initial levels of regulatory self-efficacy and strategy use but also their developmental rates of change over time. Notably, they observed a compensatory effect where less conscientious students exhibited steeper increases in strategy use, suggesting a "maturation boost" as they adapted to academic demands. Broadbent and Poon (2015) have demonstrated that specific SRL strategies—particularly time management, metacognitive regulation, and effort regulation—predict academic achievement in online contexts. This suggests that interventions targeted to students with skill-gaps, especially in these specific skills, may be more effective than giving them general study advice.

However, providing intensive SRL instruction to all students is resource-intensive and may be unnecessary for those who already possess strong regulatory skills. This has led researchers to explore targeted or adaptive interventions that provide support specifically to students who need it, based on self-assessment, early performance indicators, or behavioral analytics.

2.3 Timing and Types of Interventions

The timing of SRL interventions matters critically. Front-loaded interventions delivered at course start can establish productive learning patterns before students develop problematic habits (Bellhäuser et al., 2016; Jansen et al., 2019). Research on learning analytics has shown that early identification of at-risk students enables timely support (Ifenthaler & Yau, 2020; Jovanović et al., 2017). However, front-loaded interventions may feel abstract to students who have not yet encountered the regulatory challenges the course will present.

In contrast, "just-in-time" interventions delivered at moments of regulatory breakdown—when students fall behind, perform poorly, or show disengagement—may feel more relevant and motivating. Recent research by Cheng et al. (2025) highlights the potential and limitations of continuous monitoring tools. In their study, students using structured digital diaries three times a week demonstrated significant SRL gains during online learning, yet these improvements plateaued upon returning to in-person instruction. This suggests that while continuous prompting can effectively scaffold autonomy in unstructured online environments, its impact may diminish over time or become context-dependent.

Similarly, Breitwieser et al. (2022) demonstrated that the benefits of self-regulation prompts are highly volatile, building up during delivery but fading rapidly once stopped. This suggests many interventions function like a 'battery' needing constant recharging, highlighting the need for designs that produce lasting 'seeding' effects without continuous maintenance.

While Generative AI now enables highly personalized, real-time SRL support through dynamic profiling, these high-frequency interventions face critical sustainability challenges. Continuous AI-driven scaffolding can lead to 'prompt monotony' and a risk of over-reliance, where students passively accept automated answers rather than engaging in deep regulatory work (Luo et al., 2026). This highlights a potential advantage for lower-frequency, targeted interventions that mitigate user fatigue.

This suggests potential value in combining intervention types: early interventions that establish regulatory frameworks and later interventions that provide corrective support at critical moments. Yet few studies have examined how such combined approaches interact, or whether early interventions create conditions that enhance or diminish later intervention effectiveness.

2.4 Schedule Setting and Personal Ownership

One aspect of early interventions is fostering goal setting. Goal-setting theory (Locke & Latham, 2006) emphasizes that self-set goals create stronger commitment than assigned goals. When students create their own schedules, the concept of "being late" becomes personally meaningful—it represents a broken commitment to oneself rather than merely missing an arbitrary external deadline. This can be exemplified by a study by Theobald and colleagues (2021) which showed that introducing regular deadlines for assignments and reminding students of these deadlines did not improve distributed learning--possibly because the deadlines were not set personally by the students.

There is a persistent gap between the initial goal-setting phase and the subsequent 'goal striving' behaviors required for achievement, noting that static prompts alone are insufficient without mechanisms to support ongoing execution (Martins van Jaarsveld et al., 2025 a). This underscores the need for interventions that do not merely digitize planning but actively intervene during the performance phase. Recent attempts to scale goal-setting interventions have leveraged conversational agents to guide students through this process. Martins Van Jaarsveld et al. (2025 b) found that while providing guidance and adaptive feedback significantly improved the quality of students' initial goals (e.g., making them more specific and measurable), it failed to improve actual goal attainment. Their findings suggest a critical disconnect between the cognitive phase of setting a goal and the behavioral phase of striving to achieve it. Furthermore, the capacity to

bridge this gap is heavily mediated by internal psychological resources. Heikkinen et al. (2024) identified self-efficacy beliefs as a decisive factor in this process; their sequence analysis revealed that students with high self-efficacy employed significantly more varied and adaptive learning tactics to pursue their goals compared to their lower self-efficacy peers. Together, these studies highlight that while external guidance can improve the structure of a plan, the execution depends on internal beliefs that may require distinct behavioral triggers to activate.

This context may create particularly favorable conditions for intervention. Being notified of lateness relative to one's own plan may trigger stronger metacognitive reflection than being told one missed an externally imposed deadline. Additionally, self-scheduling requires students to engage in forethought and planning at course start, potentially priming them to be more receptive to regulatory support (Black & Deci, 2000).

Another approach to foster SRL is through providing individual feedback. According to Hattie and Timperley (2007), feedback can address four different levels: task accomplishment, processing of the task, self-regulation, and the self as a person. In the context of our study, mainly feedback about the processing of the task is relevant. As Wäschle et al. (2014) showed, it is possible to help students overcoming procrastination by providing them visualisations of their learning behavior. Similarly, Mejeh and colleagues (2024) provided timely, personalized adaptive feedback based on logfile data which resulted in positive development in several components of SRL. Relying on self-reported data, Bellhäuser et al. (2023) implemented feedback within a digital learning diary which improved goal setting, planning, adherence to schedules, and self-efficacy.

2.5 Behavioral Measures of Self-Regulation

Traditional SRL research has relied heavily on self-report instruments to measure regulatory skills and strategies. While valuable, self-reports are subject to social desirability bias, limited metacognitive awareness, and retrospective distortion (Winne & Perry, 2000). The advent of learning analytics enables complementary behavioral measurement through analysis of timestamped learning platform interactions. However, recent research cautions against relying solely on behavioral traces to capture the full spectrum of SRL. Song et al. (2025) utilized concise, 'just-in-time' daily surveys to validate trace data across multidisciplinary contexts. They found that while trace data accurately detected the engagement phase of learning, it failed to reliably capture the internal metacognitive processes of planning and reflection. Their findings underscore the necessity of retaining short, frequent self-report measures to capture the 'invisible' cognitive work that digital logs often miss.

Sequence analysis of learning behaviors offers a powerful lens to the understanding of SRL processes and the unfolding of learning events (Saqr, López-Pernas, Helske, et al., 2024). Sequence analysis reveals the patterns of transitions between learning states—such as whether students check assessment criteria before or after reviewing learning resources, or whether they seek feedback and then revise their work. In other words, sequence analysis charts the order and organization of learning events which can reveal students' learning processes in ways self-reports cannot capture (Saint et al., 2020). While sequence analysis is valuable, it does not map the relations between events and the overall structure of the learning process given that sequence analysis is strictly linear which necessitates augmenting it with network analysis methods.

Network analysis models learning behaviors as graphs in which nodes represent learning states and edges represent the relationships between them, such as co-occurrences, transitions, or associations. When the focus is on the temporal dynamics of learning and how the process unfolds over time, temporal networks are particularly useful. Network metrics, such as betweenness centrality, can be used to identify important learning events, for example by revealing which learning activities function as regulatory "hubs" that connect different aspects of the learning process (Gašević et al., 2016; Malmberg et al., 2017). In this study, we employ Transition Network Analysis (TNA) (Saqr, López-Pernas, Törmänen, et al., 2024) to capture the temporal dynamics of the learning process, modeling it as a directed network. In TNA, learners' behaviors, strategies, or cognitive states are represented as nodes, while transitions between them are represented as directed edges weighted by the frequency or probability of moving from one event to the next.

TNA offers a rich framework for visualizing, modeling and verifying the resulting process models. TNA integrates a rich set of confirmatory statistical tests to ensure the validity, stability, and interpretability of the resulting insights (Saqr, López-Pernas, Törmänen, et al., 2024). TNA includes reliability testing for overall model consistency to help verify that the constructed transition network genuinely reflects the underlying learning dynamics. To assess the strength and replicability of individual edges, i.e., the transitions between learning states, TNA uses bootstrapping to evaluate how consistently transitions appear and to discard spurious edges that do not replicate across samples. In doing so, TNA ensures that the inferences about the learning process are statistically meaningful. Finally, permutation testing (a non-parametric resampling method) is used for process comparisons, such as evaluating whether transition structures differ significantly between groups.

2.6 Intervention as Developmental Seeding

We propose that SRL interventions in online courses may operate through two distinct mechanisms. The immediate correction mechanism involves direct behavioral changes shortly after intervention delivery—students receive guidance and immediately alter their actions. Most intervention research focuses exclusively on this mechanism (Azevedo & Hadwin, 2005; Dignath & Büttner, 2008).

Recent experimental evidence supports the value of delayed effects. Tuero et al. (2022) found that SRL training yielded larger benefits at a three-month follow-up than immediately post-intervention. This suggests interventions may function as 'seeds,' initiating developmental trajectories that emerge only after students have internalized new regulatory strategies.

However, we hypothesize a second longitudinal seeding mechanism: interventions may alter students' developmental trajectories such that differences emerge over time even if immediate effects are limited. Early interventions might establish metacognitive frameworks, regulatory habits, or strategic dispositions that shape how students respond to later course demands and subsequent interventions. This mechanism suggests that intervention effects should be assessed longitudinally rather than only immediately post-intervention.

Our dual-intervention design enables us to test this framework. If the initial survey intervention operates primarily through immediate correction, we should observe behavioral changes shortly after its delivery. If it also operates through longitudinal seeding, we should observe that experimental and students in the control group have diverged in their learning patterns by the time the LoS intervention is delivered, even if immediate post-survey differences were limited.

3 Materials and Methods

3.1 Educational Context and Course Design

The study was conducted within an asynchronous online Project Management course for first-year business students at a Finnish University of Applied Sciences. A defining pedagogical feature was the requirement for students to set personal learning goals and schedules at the onset of the course. The numerical goal (scale: 1–5) determined the workload; for example, the lowest passing grade required completing four assignments, while the highest grade required eleven.

The course utilized a self-scheduling mechanism. As their first assignment, students submitted a personal schedule detailing expected submission dates for their assignments. This design meant that lateness was defined by students' own commitments rather than instructor-imposed deadlines, making the concept of schedule adherence personally meaningful.

3.2 Participants and Experimental Design

The study employed a randomized controlled trial design. The initial cohort consisted of N=271 students randomly assigned to a Control group (n=133) and an Experimental group (n=138) within the Learning Management System (LMS). Due to the longitudinal nature of the study and the requirement to analyze pre- and post-intervention behaviors, specific subsamples were constructed for analysis:

1. Baseline Sample (N=271): Used to check for randomization and baseline equivalence.
2. Survey Intervention Sample (N=206): Used to analyze the effects of the early-stage support, comparing all active Control and Experimental group students.
3. LoS Intervention Sample (N=106): A matched-pairs sample comparing students who received the LoS intervention (Experimental) against a matched Control group who exhibited similar delay patterns but received no intervention.

3.3 Procedure and Interventions

The study utilized a sequential, two-stage intervention strategy designed to support different phases of the SRL cycle.

3.3.1 Initial Survey Intervention (Early-Course)

At the beginning of the course, all students completed a baseline SRL survey adapted from validated instruments (Barnard et al., 2009; Broadbent et al., 2022; Jansen et al., 2017). The survey assessed skills such as asking for help from the teacher (Askhelp), reviewing assessment criteria (Criteria), setting realistic deadlines (Deadlines) and goals (Goals), thinking about learning needs before starting a task (Prethink), scheduling coursework (Schedule), doing self-assessment (Self-Assessment), setting standards for coursework (Standards), considering

different strategies (Strategies), summarizing learning content (Summary), choosing a suitable time (Time) and location (Location) for learning, and managing distractions (Distractions).

Following the logic established by (Broadbent et al., 2022), students in the experimental group who reported low self-assessment scores (below the mid-point on specific sub-scales) received immediate, targeted feedback prompts. These prompts provided specific instructional strategies to improve the identified weak areas (e.g., specific techniques for time management or criteria analysis). For the subscales from Broadbent et al (2022), we used the original interventional prompts suggested by the authors. For the other subscales adopted (Barnard et al., 2009; Jansen et al., 2017) we drafted prompts with a similar structure. An example of the prompt is shown in Appendix A. Students in the control group completed the survey but received no specific feedback.

3.3.2 LoS Intervention (Mid-Course)

The second intervention targeted regulatory problems during the performance phase. The course instructor systematically monitored student progress against the personal schedules submitted at the start of the course.

- Experimental Condition: When a student in the experimental group fell behind their self-determined schedule, they received a personalized email notification. This communication noted the specific delay relative to their plan and asked them to revise their schedule.
- Control Condition: Students in the control group who fell behind their schedules received standard course communications (e.g., general announcements) but no personalized alerts regarding their lateness. To enable rigorous behavioral comparison, "virtual" intervention timestamps were assigned to students in the control group, matching the moment they would have received a notification based on their delay patterns.

3.4 Data Collection and Measures

3.4.1 Self-Reported Measures

Longitudinal data on student perceptions were collected via study diaries submitted alongside course assignments. These diaries measured variables explained in **3.3.1 Initial Survey Intervention (Early-Course)** using single-item measures to reduce response burden while allowing for repeated measurement. The statements of the diary are presented in Appendix B.

The study utilized a repeated-measures design using single-item indicators to assess SRL factors. This method addresses the "response burden" challenge identified in longitudinal research (Fisher et al., 2016), where the length of traditional multi-item scales can lead to attrition or low-quality data. By utilizing concise measures, we enabled real-time reporting of skill development (Solhan et al., 2009), thereby mitigating the recall bias common in retrospective assessments.

While single-item measures have historically faced scrutiny regarding validity, contemporary evidence supports their feasibility and reliability, particularly for real-time monitoring (Allen et al., 2022). This approach mirrors the methodology of Saqr and López-Pernas (Saqr & López-Pernas, 2024), leveraging high-frequency sampling to capture dense longitudinal data. This structure ensures statistical adequacy for analyzing intra-individual variability, even within

smaller cohorts (Maas & Hox, 2005; Martin et al., 2015; Raudenbush & Bryk, 2002). This approach is further supported by Song et al. (2025), who demonstrated that concise daily surveys are essential for capturing planning and reflection phases that remain invisible to behavioral tracking.

3.4.2 Behavioral Trace Data

Student interactions with the Moodle LMS were captured in detailed logs, resulting in a dataset of timestamped activities. To facilitate meaningful sequence analysis, raw log entries were coded (Molenaar & Chiu, 2014) into eight high-level learning states following the coding scheme used in previous work of Heikkinen, Bellhäuser, et al. (2025). The detailed recoding scheme is displayed in Appendix C.

- Assessment Criteria (AC): Viewing rubrics or requirements.
- Assignment Viewed (AV): Reading assignment instructions.
- Course Page (CP): Navigating the main course interface.
- Evaluating (EV) (Teacher action): Teacher performs the evaluation of an assignment. This state captures the student's engagement with the teacher's direct evaluation activity. When a teacher grades a submission, the student is notified; this node therefore represents the student's reaction to this external instructional action.
- Feedback (FB): Viewing grades or instructor feedback.
- Learning Resource Viewed (LR): Accessing video lectures or learning materials.
- Self-Evaluation (SE): Submitting self-reflection forms.
- Submitting (Sub): Uploading assignments.

Data preprocessing involved identifying "learning sessions" to segment the continuous stream of actions. A session was defined as a sequence of activities separated by a temporal gap of at least 30 minutes (Jovanović et al., 2017). In addition to sequence composition, we calculated the temporal investment for each learning state. We computed the duration of each activity by calculating the time difference between consecutive log timestamps. To account for 'idle time' or students leaving the platform open, we applied a cap to the duration of single actions, excluding outliers exceeding 30 minutes from the aggregate time-on-task analysis.

3.5 Data Analysis

We adopted a mixed-methods analytical approach, integrating statistical analysis of self-reports with advanced process mining via TNA (Saqr, López-Pernas, Törmänen, et al., 2024) of behavioral traces. The analysis was structured to address the four specific research questions as follows.

3.5.1 RQ1: Immediate Effects of the Initial Survey Intervention

To rigorously assess the baseline comparability of the Control and Experimental groups, we moved beyond standard null-hypothesis significance testing. A non-significant result in a traditional difference test (e.g., Student’s t-test) indicates a failure to reject the null hypothesis but does not statistically confirm that the groups are equivalent. Therefore, we employed the Two One-Sided Tests (TOST) procedure to test for equivalence.

We utilized the TOSTER package in R (Lakens, 2017) to determine if the observed effect size between groups fell within a specified equivalence bound. For this study, we established the smallest effect size of interest (SESOI) at Cohen's d = 0.3. This procedure tests the null hypothesis that the difference between group means is large (i.e., outside the equivalence bounds) against the alternative hypothesis that the difference is small enough to be considered equivalent.

To determine whether the initial, personalized SRL feedback altered students' immediate perceptions and processes, we employed two complementary analyses: self-report analysis with 2x2 ANOVA and permutation tests on transitions. We conducted a series of 2x2 ANOVA on the diary survey data collected with the next assignment submission following the intervention. The model included Group (Control vs. Experimental) and Time (Pre- vs. Post-intervention) as factors. A simple comparison of post-test scores would fail to account for baseline differences. By focusing on the Interaction Effect (Group x Time), we could isolate the specific variance attributable to the intervention, separating it from the general "maturation effects" observed as students naturally progress through a course. To assess immediate behavioral changes, we utilized permutation tests (1000 permutations) on the transition matrices derived from log data immediately following the survey. We hypothesized that the intervention might not immediately increase activity volume but would alter the structure of learning. Permutation tests allowed us to identify specific behavioral links (e.g., Feedback -> AssignmentViewed) that were significantly strengthened by the intervention, indicating immediate responsiveness to the feedback prompts. To interpret significant Interaction Effects (Group x Time), we conducted a simple main effects analysis. This involved performing pairwise comparisons (paired sample t-tests) for each group separately to determine if the change from pre- to post-intervention was statistically significant within the Experimental group versus the Control group.

3.5.2 RQ2: Longitudinal Seeding Effects

To investigate whether the initial intervention altered developmental trajectories before the second intervention occurred (the "seeding effect"), we analyzed the baseline behaviors of the LoS sample. We compared the behavioral states of the Experimental and Control groups at the moment they triggered the LoS criteria, but before the notification was sent. We utilized Mosaic plots and standardized residuals from Chi-square tests to visualize distributional differences. If the groups displayed significant behavioral differences prior to the second intervention (despite being matched on lateness), it would provide evidence that the initial survey intervention initiated a long-term divergence in learning strategies that persisted over weeks.

3.5.3 RQ3: Effects of the LoS Intervention

To examine whether the LoS notifications triggered a "metacognitive awakening" and behavioral correction for students falling behind, we focused on the divergence between matched pairs using a 2x2 ANOVA and a transition network analysis. We analyzed the self-report variables using the 2x2 ANOVA framework described above to examine changes in how students perceive their learning. Using the TNA package (Tikka et al., 2025), we mined frequent process patterns (e.g., Feedback -> SelfEvaluation) and calculated their Support (the proportion of students exhibiting the pattern). Simple activity counts (e.g., "number of clicks") fail to capture self-regulatory loops. By analyzing subsequences, we could determine if the intervention successfully prompted recursive regulatory cycles rather than just linear task completion. We used Chi-

squared tests with Bonferroni corrections to ensure that observed differences in pattern frequency were statistically robust.

3.5.4 RQ4: Behavioral Patterns and Trade-offs

To characterize the architecture of the learning process and identify trade-offs (e.g., sacrificing content consumption for regulation), we employed TNA (Saqr, López-Pernas, Törmänen, et al., 2024). TNA was selected over other methods given rich modeling and extensive suite of confirmatory testing—reliability testing, bootstrapping of edges, case-dropping stability analysis, and permutation testing for differences to confirm that observed patterns and key transitions in temporal learning networks are meaningful and not merely artifacts of the data or analytical choices.

We constructed directed graphs where nodes represented learning states (e.g., Learning Resource Viewed, Self-Evaluation) and edges represented weighted directed transitions. We visualized these transition models to reveal the topology of the learning process. To visualize these structural differences, we constructed subtraction TNA networks. In that, we subtracted the transition probabilities of the Control group from those of the Experimental group. Edges were colored to indicate the direction of the shift: green edges represented transitions that became significantly more frequent in the Experimental group (positive difference), while red edges indicated transitions that became significantly more prevalent in the Control group (negative difference), based on the significance thresholds established by the permutation tests.

We calculated Betweenness Centrality for each node in the network. Centrality metrics reveal the role of specific activities. High Betweenness for regulatory nodes would indicate that they act as "hubs" connecting different learning phases, whereas low Betweenness would suggest isolated, siloed behavior. To explore possible trade-offs, we compared specific edge weights (e.g., CoursePage -> LearningResource) between groups. This allowed us to quantify significant differences between the Experiment and Control groups. To evaluate variations in sequential learning behaviors between the experimental and control cohorts, we utilized the TNA package (Saqr, López-Pernas, Törmänen, et al., 2024). We extracted subsequences with a maximum length of five actions and calculated their frequency (total pattern counts). To determine which subsequences statistically differentiated the two groups, we conducted Chi-squared tests adjusted with the Bonferroni correction to control for Type I errors. The significant patterns were then visualized to illustrate the distributional differences between the cohorts.

Finally, we specifically examined transitions originating from the 'Evaluating' state. As this state represents the student actions right after the teacher has graded and given feedback for the student. Outgoing transitions from this node serve as indicators of how students react to instructor intervention. We analyzed whether students responded to teacher evaluation by reviewing feedback (Feedback), checking requirements (Assessment Criteria), reviewing instructions (Assignment Viewed), deepening theoretical understanding (Learning Resource Viewed), or engaging in reflection (Self-Evaluation).

4 Results

4.1 Baseline Data Analysis

Before examining intervention effects, we tested for equivalence between groups. While the C and E groups had statistically similar overall activity counts (C: M=198.24, SD=241.4; E: M=234.24, SD=293.12; p=0.27), a TOSTER equivalence test failed to confirm statistical equivalence (p=0.0862) on effect size 0.3. This suggests the groups, though not significantly different, may have had small baseline behavioral differences. To ensure these baseline levels did not confound our findings, we utilized a mixed-model approach focusing on the Group × Time interaction effect. This approach identifies divergent developmental trajectories between the groups, effectively treating each participant as their own control and isolating the impact of the intervention from initial state variances.

4.2 Immediate Effects of the Initial Survey Intervention

4.2.1 Survey Results (ANOVA)

Self-reported data (Table 1) revealed a significant interaction effect for 'Criteria' ($F(1,199)=5.62$, $p=.019$), suggesting the initial survey intervention successfully altered how the Experimental group engaged with assignment criteria compared to the Control group (Table 1). Students in both groups showed significant changes over time on the following variables: Deadlines ($p < .001$), AskHelp ($p < .01$), Schedule ($p < .05$), and Goals ($p < .05$). This is likely a general "maturation" effect from simply being in the course.

Table 1: Interaction Effects (Survey Intervention)

Variable	Group F	Group p	Group ηG2	Time F	Time p	Time ηG2	Interaction F	Interaction p	Interaction ηG2
SAndLSchedule	0.14	.710	.001	4.36	**.038**	.005	0.61	.435	.001
Deadlines	0.00	.985	.000	13.10	**<.001**	.018	1.37	.243	.002
Standards	1.27	.260	.005	0.13	.723	.000	1.64	.203	.002
Schedule	0.08	.775	.000	6.58	**.011**	.005	0.00	.974	.000
Prethink	1.46	.228	.006	1.55	.215	.002	1.40	.238	.001
Strategies	0.00	.999	.000	2.49	.116	.002	0.04	.849	.000
No Distraction	0.70	.403	.003	3.22	.074	.003	0.00	.995	.000
Time	1.92	.167	.008	0.01	.914	.000	0.14	.705	.000
Location	0.22	.637	.001	0.88	.348	.001	0.41	.523	.000
Criteria	**0.00**	**.978**	**.000**	**0.01**	**.911**	**.000**	**5.62**	**.019**	**.006**
Summary	0.46	.498	.002	0.78	.377	.000	0.92	.340	.001
AskHelp	0.02	.896	.000	7.20	**.008**	.005	3.46	.064	.003
SelfAssess	0.18	.675	.001	0.04	.851	.000	0.02	.900	.000

Note: Significant p-values are highlighted in bold. Degrees of freedom (df) are 1, 199 for most variables.

Table 2 details the significant interaction observed for the 'Criteria' variable ($F(1, 199) = 5.62$, $p = .019$). The interaction effect appears to be driven by divergent trends between the two groups. While not statistically significant at the .05 level in isolation, the Experimental group showed a trend toward improved self-evaluation of criteria (Estimate = -0.37, $p = .081$), whereas the Control group showed a non-significant decline (Estimate = 0.34, $p = .111$). The statistical significance of the interaction confirms that these opposing trajectories—the Experimental group improving while the Control group declined—were distinct from one another.

Table 2: Simple Effects Analysis for the 'Criteria' Interaction

Comparison	*Contrast*	*Estimate*	*SE*	*t-ratio*	*p-value*
Simple Effects					

Control Group	*Before - After*	*0.34*	*0.21*	*1.60*	*.111*
Experimental Group	*Before - After*	*-0.37*	*0.21*	*-1.75*	*.081*
Between-Group Effects					
At 'Before'	*Control - Experimental*	*0.36*	*0.34*	*1.06*	*.293*
At 'After'	*Control - Experimental*	*-0.35*	*0.33*	*-1.05*	*.294*

4.2.2 Process Analysis

Students spent the most significant proportion of their time on "Submitting" and "Assessment Criteria", indicating that assessment-related activities were the primary drivers of time-on-task (Appendix D). Behavioral analysis corroborated this. The results of permutation tests (1,000 permutations) comparing transition probabilities between the Experimental and Control groups showed that the experimental group showed significantly higher Feedback→Assignment Viewed transitions (0.11 relative support), indicating that the intervention, which focused on how to improve SRL skills, made the students focus on the relevant assignment.

4.3 Longitudinal Seeding: Behavioral Divergence Prior to LoS

We compared the absolute frequency of states before and after intervention. While both groups increased their total activity volume over time (a natural course progression), the distribution of this increase differs (Appendix E). To investigate the longitudinal seeding hypothesis (RQ2), we visualized the behavioral distributions of the matched sample using mosaic plots of standardized residuals (Figure 1). The left panel illustrates behaviors before the LoS intervention—specifically when students fell behind schedule but prior to receiving the notification. Although these students were matched based on their delay patterns, the analysis reveals a statistically significant divergence ($chi^2_7 = 624.9$, $p < .001$).

This confirms that the initial survey intervention initiated a delayed drift in learning behaviors that manifested weeks into the course, even before the second intervention occurred. The right panel captures the behavioral shifts following the LoS intervention ($chi^2_7 = 282$, $p < .001$). This demonstrates that the behavioral distinctiveness of the Experimental group was not only maintained but reinforced after receiving the LoS alerts, characterized by a unique distributional profile of regulatory activities.

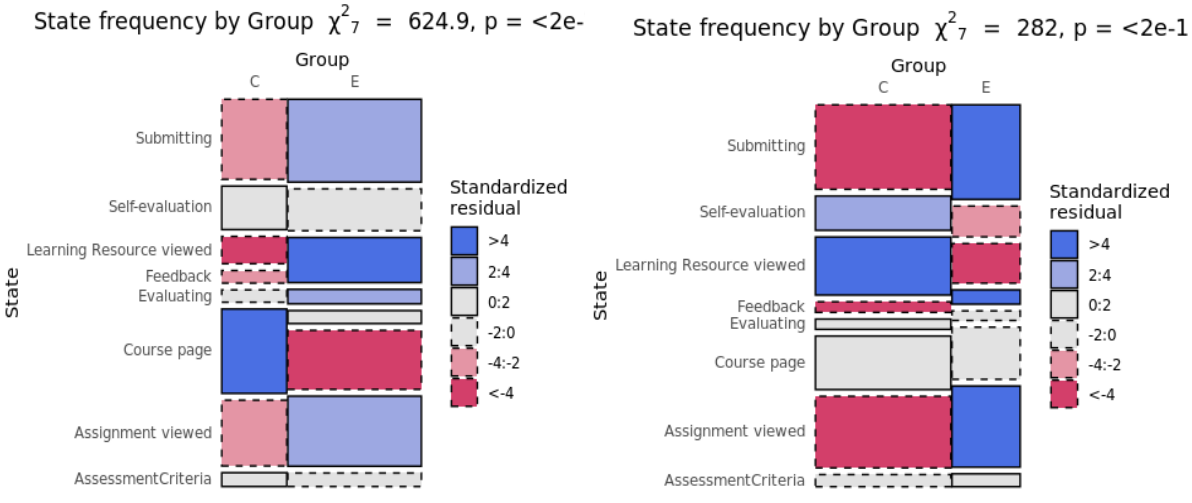


Figure 1. Mosaic plots comparing behavioral distributions between Control (C) and Experimental (E) groups before and after the LoS intervention. The **width of each column** represents the relative activity volume (total number of log actions) contributed by that group during the specified period. The relative narrowing of the Experimental column in the "after" panel reflects the cohort's shift toward more targeted, lower-frequency regulatory actions compared to the Control group. Tile colors represent standardized residuals: **blue** indicates observed frequencies significantly higher than expected (residual > 2); **red** indicates frequencies significantly lower than expected (residual < -2).

4.4 Effects of the LoS Intervention

4.4.1 Survey Results (ANOVA)

Table 3 summarizes the 2x2 ANOVA results across all self-reported variables. While significant main effects of Time were observed for both 'Goals' ($p < .001$) and 'Deadlines' ($p = .001$)—indicating a general shift across the entire sample—the critical finding was a significant Group × Time interaction for 'Deadlines' ($p = .001$), which confirmed that the intervention group's trajectory was uniquely altered. This counter-intuitive finding suggests the intervention made the E-group less overconfident and more realistic about their goal-setting and time management. This indicates that all students, regardless of group, adjusted their planning behaviors as the course progressed, likely due to course maturation. However, the interaction effects isolate where the intervention specifically altered this trajectory.

Table 3: Summary of 2x2 ANOVA on Self-Reported Variables

Variable	Intervention Effect	Time Effect	Interaction Effect
	F, p, ηG2	F, p, ηG2	F, p, ηG2
Goals	0.14, .707, .001	13.50, <.001***, .021	2.96, .088, .005
Deadlines	0.17, .683, .001	11.60, .001**, .026	11.00, .001**, .027
Standards	0.57, .451, .005	0.17, .682, .000	1.34, .249, .002
Schedule	2.49, .118, .019	5.57, .020*, .011	1.09, .300, .002

Prethink	0.06, .811, .000	0.21, .645, .000	0.05, .823, .000
Strategies	0.76, .384, .006	3.29, .073, .006	0.01, .930, .000
Time	3.12, .080, .023	1.66, .201, .004	0.75, .389, .002
Location	2.06, .154, .016	0.11, .746, .000	0.56, .456, .001
Criteria	0.24, .625, .002	0.06, .814, .000	0.42, .521, .001
Summary	0.03, .858, .000	0.44, .508, .001	0.02, .885, .000
Askhelp	0.14, .705, .001	<0.01, .964, .000	2.01, .160, .003
Self-Assessment	0.01, .940, .000	0.96, .331, .002	0.04, .849, .000

Note. F-statistics are reported with (1, 101) degrees of freedom. Effect size is measured using Generalized Eta-Squared (ηG2). The "Time Management" variable corresponds to "Time" in the source data to avoid confusion with the "Time Effect" column. *$p < .05$. **$p < .01$. ***$p < .001$.

Table 4 deconstructs the significant interaction for "Deadlines." The Simple Effects analysis reveals that the shift was driven entirely by the Experimental group, who showed a highly significant change from 'Before' to 'After' ($t = 4.99$, $p < .001$). In contrast, the Control group's deadline perception remained statistically stagnant ($p = .951$). This confirms that the intervention specifically catalyzed a re-evaluation of deadlines in the experimental group, rather than the change being a general course artifact.

Table 4: Simple Effects Analysis for the 'Deadlines' Interaction

Comparison	Contrast	Estimate	SE	t-ratio	p-value
Simple Effects					
Experimental Group	Before - After	1.59	0.32	4.99	< .001
Control Group	Before - After	0.02	0.35	0.06	.951
Between-Group Effects					
At 'Before'	Experimental - Control	0.61	0.46	1.35	.180
At 'After'	Experimental - Control	-0.95	0.50	-1.91	.059

4.4.2 Process Analysis

To explore the effect of the LoS intervention, we mapped the significant changes in behavioral transitions for the Experimental group (Appendix F). The permutation tests (1,000 permutations), identified transitions that significantly increased or decreased in probability. The strongest shift occurred in the transition from Course Page-> Learning Resource Viewed (+0.18), indicating a major increase in content access. Simultaneously, new regulatory loops emerged connecting Learning Resources -> Self-evaluation (+0.06) and Feedback -> Assignment Viewed (+0.06), while the immediate path from Self-evaluation -> Submitting significantly decreased (-0.08).

We were not merely interested in the probabilities of single-step transitions. While transition probabilities capture local dependencies, they often fail to reflect the strategic intentionality or the cyclical nature of self-regulated learning. Instead, we sought to identify recurring sub-sequences that represent distinct study episodes. Analyzing these patterns is essential because it allows us to distinguish between fragmented navigation and cohesive regulatory loops (e.g., a student moving systematically between Assessment Criteria, Self-evaluation, and Content). Identifying these frequent sub-sequences provides a higher-fidelity view of how the intervention altered the structure of students' learning episodes, rather than just the frequency of their clicks. Figure 2 illustrates the significant differences in learning pattern frequencies between the two groups. Bonferroni-corrected analysis identified eleven significant patterns ($chi^2 = 11.51–53.72$, $p < 0.05$). As shown in the plot, the Experimental group exhibited a distinct dominance in regulatory loops. Specifically, the Feedback -> Self-evaluation pattern was 6.75 times more prevalent in the Experimental group compared to the Control group (108 vs. 16; $chi^2 = 53.72$, $p < 0.001$), while the Self-evaluation -> Feedback pattern showed a ninefold increase (45 vs. 5; $chi^2 = 24.77$, $p < 0.001$). These visual and statistical results highlight a qualitative shift in the Experimental group towards systematic feedback integration and active self-assessment.

The Experimental group demonstrated a substantial shift toward recursive self-assessment, with repetitive self-evaluation patterns (SE^2 and SE^3) occurring significantly more often than in the Control condition (see Figure 2). However, this increased regulation appeared to replace prolonged content study. The plot reveals that the Control group maintained higher fidelity to extended learning resource chains (LR^4 and LR^5), highlighting a divergence where the Experimental group favored variety and regulation over the extended content immersion seen in the Control group.

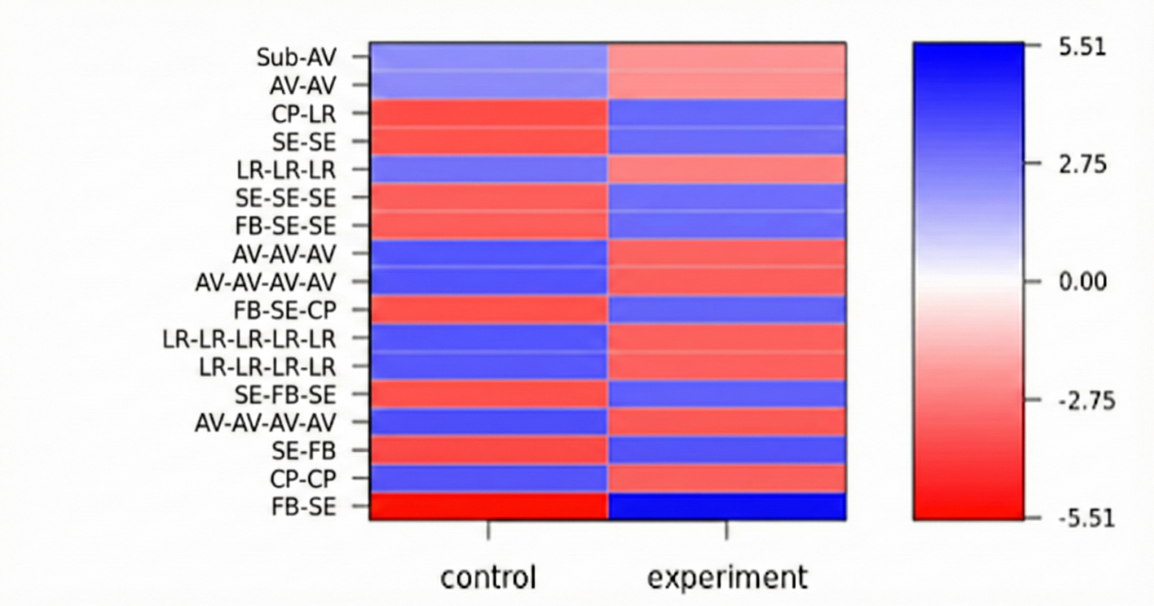


Figure 2. Heatmap of the most frequent two-state behavioral sequences. The color scale and numeric values represent the standardized residuals (z) comparing sequence frequencies between groups. Positive values (blue) indicate sequences that are significantly more prevalent in the Experimental group, while negative values (red) indicate sequences more prevalent in the

Control group. For instance, the sequence FB-SE (z = 5.51) represents a highly significant "regulatory loop" unique to the intervention cohort.

Figure 3 visualizes the structural transformation in learning dynamics by comparing transition probabilities between the Control and Experimental groups. The contrast between the two panels illustrates a complete reversal of behavioral dominance.

The Left Panel (Before LoS Intervention) is dominated by red edges, indicating transitions that were significantly more frequent in the Control group. The most prominent feature is the strong negative difference in the Course Page -> Learning Resource transition (-0.31). This confirms that prior to the second intervention, the Control group was far more likely to transition from the course home page directly into learning content. Additionally, the Control group showed stronger connections transitioning out of content into evaluation activities, as seen in the negative weights for Learning Resource -> Self-evaluation (-0.09).

The Right Panel (After LoS Intervention) reveals a dramatic shift. The intervention triggered a surge in specific pathways for the Experimental group, indicated by green edges. Notably, the Course Page -> Learning Resource transition shifted to positive (+0.18), suggesting students in the experimental group began accessing learning materials more frequently from the course menu than the Control group. Furthermore, a new link between receiving evaluation from the teacher and submitting an assignment emerged (+0.07). Simultaneously, the direct path from Self-evaluation -> Submitting showed a negative difference (-0.08), indicating that students in the experimental group were less likely to submit immediately after reflecting, potentially engaging in further revisions. This confirms that the LoS notification successfully 'flipped' the learning behavior; students in the experimental group not only returned to content more frequently but actively connected that content to self-reflection and submission.

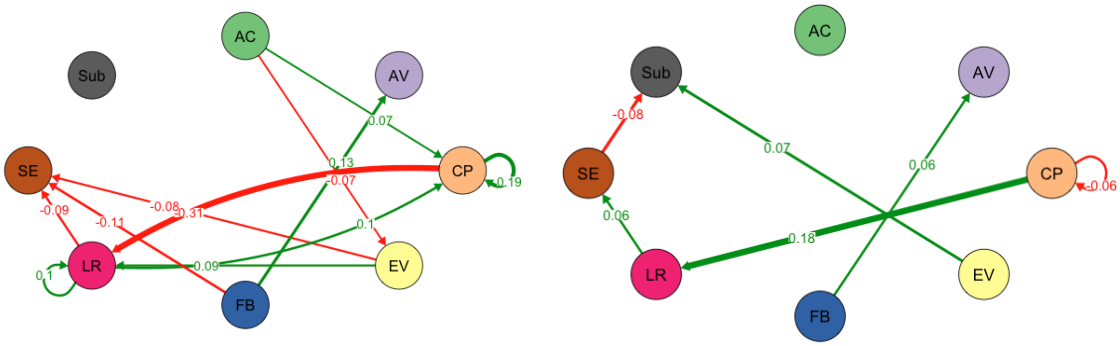


Figure 3. Network graphs showing statistically significant differences ($p < .05$) in transition frequencies based on permutation tests. Green edges indicate a significant increase in transitions between states, while red edges indicate a significant decrease. The comparisons are showing the differences between Control vs. Experimental before (left) and after (right) the LoS intervention.

We analyzed Betweenness Centrality to identify which learning states served as central "hubs" in the process flow, connecting different phases of learning (Appendix G). In the pre-intervention phase (Left Panel), the Experimental group already displayed a more distributed network structure, with high betweenness centrality values across Feedback, Evaluating, and Assessment Criteria. In contrast, the Control group showed generally lower centrality scores, with Assessment Criteria serving as their primary, albeit less dominant, structural hub. Following the intervention (Right Panel), Feedback emerged as the most critical node for both groups, acting as the primary bridge in the network. However, the groups diverged significantly in how this feedback was integrated. For the Experimental group, high centrality values persisted for Evaluating and Assessment Criteria. This indicates a resilient structure where the teacher's evaluation activity was not an isolated event but a central hub that bridged different learning phases. Rather than ignoring the teacher's grading action, students in the experimental group used it as a pivot point to transition into checking criteria and engaging in self-evaluation. Conversely, for the Control group, the surge in Feedback centrality dominated the topology. Most strikingly, the Control group exhibited near-zero betweenness values for Self-evaluation and Assignment Viewed. This structural disconnection suggests that while students in the control group accessed feedback, they were unable to use it as a bridge to regulatory actions (self-evaluation) or clarifying steps (viewing assignment instructions), effectively leaving these critical activities isolated from the feedback loop.

5 Discussion

5.1 The Longitudinal Seeding Effect

A critical finding of this study is the divergence of the Control and Experimental groups prior to the second intervention. While recent longitudinal research by Kryshko et al. (2026) suggests that motivational regulation strategies develop naturally over time through a "maturation boost"—particularly in less conscientious students —our findings indicate that this developmental process need not be left to chance. Kryshko et al. (2026) observed that students' strategy use evolved linearly or remained stable depending on personality traits. In contrast, our "seeding effect" demonstrates that targeted early interventions can artificially induce a divergence in developmental trajectories. This suggests that external scaffolding can effectively simulate the "maturation" process, creating productive regulatory habits in students who might otherwise lack the dispositional traits (such as high conscientiousness) to develop them spontaneously.

Our results challenge the volatility observed by Breitwieser et al. (2022), where benefits vanished immediately after prompting ceased. Conversely, our initial intervention created a 'seeding' effect, with group differences emerging weeks later. This divergence, captured by the significant Group × Time interaction, demonstrates that the intervention altered the developmental trajectory of the students rather than merely magnifying pre-existing baseline characteristics. By focusing on these interaction effects, we isolate the unique growth patterns triggered by the 'seed' from initial behavioral states. This suggests that while repetitive prompting acts as a temporary scaffold, targeted early reflection can fundamentally recalibrate the learning path, enabling autonomous growth.

Our observed 'seeding effect' aligns with Tuero et al. (2022), who found that SRL training benefits grew significantly stronger at a three-month follow-up compared to immediate post-tests. This corroborates our hypothesis that effective interventions do not merely produce transient behavioral spikes but alter long-term developmental trajectories, initiating regulatory growth that becomes more visible over time.

5.2 Metacognitive Awakening Mechanism

The experimental group's increased engagement with self-regulatory behaviors—specifically the increase in Feedback-SE patterns—aligns with the ANOVA findings regarding "Deadlines". The significant drop in self-reported deadline confidence suggests the notifications triggered a "metacognitive awakening". By explicitly highlighting the gap between planned and actual progress, the notification likely stripped away overconfidence, prompting students to evaluate their learning process rather than continuing on autopilot. Our observation that the LoS alert triggered a realistic reassessment of deadlines aligns with recent findings by Wang et al. (2024). In their experimental study, providing students with real-time feedback on their monitoring accuracy significantly reduced overconfidence, shifting students towards a more realistic self-assessment. Similarly, our intervention likely functioned as a 'reality check', piercing the optimism bias often inherent in student planning and forcing a recalibration of their progress against objective criteria.

Our findings regarding the power of 'just-in-time' alerts align with recent distinctions made by Li et al. (2025) between process-based and condition-based scaffolding. They found that AI-driven prompts based solely on SRL processes were ineffective; scaffolding only improved regulation when it accounted for the learner's dynamic conditions (e.g., prior knowledge, resource awareness). Similarly, our LoS intervention was effective precisely because it was condition-dependent—it was not a generic reminder to 'work faster' but a specific alert triggered by the discrepancy between the student's *current state* and their *self-determined plan*. This suggests that the 'condition'—in this case, the violation of a personal commitment—is the critical ingredient that transforms a notification into a regulatory catalyst.

The mechanism driving this awakening appears to be the forced confrontation with one's own progress. This interpretation is supported by Theobald and Bellhäuser (2022), who found that providing students with strategy suggestions alone ('directive feedback') failed to improve self-regulation. Significant gains only occurred when students received feedback that explicitly highlighted the gap between their goals and their actual performance. Similarly, our LoS alert functioned as a high-stakes 'reality check,' supplying the critical diagnostic signal that allowed students to recognize the need for change.

The behavioral shift observed in our experimental group can be understood through the longitudinal profiles (Heikkinen, Saqr, et al., 2025). They characterize a distinct group of 'disengaged low regulators' who are at high risk of stagnation. Our findings suggest that the LoS alert functioned as a critical catalyst, effectively triggering a conversion from this passive state into 'low-regulating improvers' who actively seek feedback and engage in self-assessment. This indicates that while regulatory profiles may appear stable, they remain malleable to targeted, high-salience triggers that force a re-evaluation of learning strategies.

5.3 Trade-offs and Side Effects

The intervention triggered a strategic recalibration. However, this came with trade-offs. The Control group demonstrated higher frequencies of repetitive instructional behaviors (AV^4, AV^5), suggesting they pursued deeper engagement with specific resources. In contrast, the Experimental group's emphasis on assessment sometimes came at the expense of content engagement, highlighting the complex side effects of scaffolding in online environments.

The LoS intervention triggered what we term a "metacognitive awakening" in experimental group, evidenced by dramatic increases in regulatory behaviors: higher Feedback-SE patterns and increases in SE-Feedback loops. We propose two interrelated mechanisms explaining these effects. First, explicitly highlighting the gap between planned and actual progress prompted students to evaluate their learning process rather than continuing on autopilot—questioning not just when they completed tasks, but how well they understood the material. Critically, this effect likely depended on students having set their own deadlines; personal ownership of schedules may be a prerequisite for such metacognitive responses to lateness notifications. Second, the notification triggered a strategic recalibration of learning approaches. The results reveal that students in the experimental group significantly increased their transitions from the Course Page to Learning Resources (+0.18) after the intervention, challenging the assumption that regulation simply replaces content engagement.

However, the nature of this engagement shifted. While students in the experimental group accessed resources frequently, the significant decrease in the Self-Evaluation -> Submitting (-0.08) transition indicates they were less likely to submit immediately after reflection. Instead, the emergence of the Evaluating -> Submitting (+0.07) pathway suggests a pattern of responsiveness, where students likely utilized teacher evaluation or feedback as a trigger to finalize their work rather than submitting in isolation. This suggests that the "trade-off" was not a reduction in content access, but a shift toward regulated consultation and delayed submission. Students checked resources and integrated evaluation feedback prior to submission, rather than engaging in the linear, immediate task completion observed in the Control group.

Our findings offer a solution to the "over-regulation" effect observed in recent diary-based studies. Cheng et al. (2025) reported that while digital diaries boosted SRL during online learning, skill development stagnated during subsequent phases, likely because the repetitive weekly prompts became burdensome or redundant. In contrast, our LoS intervention functioned as a targeted 'metacognitive awakening' rather than a routine compliance task. By reserving the intervention for moments of regulatory failure (falling behind), we maintained the saliency of the feedback, preventing the habituation and burnout often associated with continuous, high-frequency self-monitoring protocols.

These findings reveal how interventions create complex redistributions of learning effort. The scheduling prompts addressed time management (the intended effect) but also altered how students allocated attention across activities. This supports the theoretical position that self-regulation benefits from external scaffolding (Azevedo & Hadwin, 2005), while demonstrating that such scaffolding produces unexpected side effects: the intervention emphasized assessment

716 at the expense of learning resources and promoted regulatory intensity at the potential cost of
717 content depth.

718 **5.4 Practical Implications**

719 For instructors and course designers, our findings suggest several actionable strategies. First,
720 implement early SRL interventions even if immediate effects seem modest—their value may lie
721 in seeding long-term developmental changes. Second, personalize interventions based on both
722 individual characteristics (low SRL skills) and behavioral indicators (falling behind schedule);
723 our dual-intervention approach suggests these complementary targeting methods address
724 different regulatory challenges. Third, give students ownership over their schedules where
725 feasible; self-determined deadlines appear to create stronger foundations for effective
726 intervention. Fourth, recognize that promoting self-regulation may reduce time spent on other
727 learning activities—interventions should be designed with awareness of these trade-offs,
728 potentially pairing regulatory prompts with guidance on maintaining balanced engagement
729 across learning activities.

730 Our findings suggest that the timing of intervention triggers is as critical as their content. While
731 continuous, high-tech scaffolding can initially boost performance, it risks saturation and 'prompt
732 monotony' over time, leading to disengagement. In contrast, interventions triggered only by
733 specific regulatory failures—such as falling behind schedule—maintain their salience. By
734 functioning as a distinct 'alarm' rather than background noise, such alerts can trigger a
735 'metacognitive awakening' without inducing the fatigue associated with constant monitoring
736 (Luo et al., 2026).

737 Finally, our design balances the tension between support and independence. De Vreugd et al.
738 (2025) caution that while students need support in setting goals, excessive structuring can
739 impinge on their autonomy. Our approach—allowing students to set their own deadlines
740 (autonomy) but intervening when they fail to meet them (support)—navigates this balance,
741 providing scaffolding that is responsive rather than restrictive (de Vreugd et al., 2025).

742 **5.5 Limitations and Future Directions**

743 Several limitations warrant consideration. Our study was conducted in a single course context
744 (business students, project management, Finnish university), which may limit generalizability.
745 The specific content and structure of our interventions—Broadbent-based SRL feedback and
746 simple schedule notifications—represent only two of many possible intervention approaches.
747 Future research should examine whether similar seeding effects and metacognitive awakenings
748 occur with different intervention designs, in different course contexts, and with different student
749 populations.

750 Additionally, while our sequence and network analyses revealed important behavioral
751 differences, we cannot definitively establish whether these changes led to better learning
752 outcomes. The experimental group's increased self-regulation occurred alongside decreased deep
753 content engagement—whether this trade-off ultimately benefits or harms learning likely depends
754 on context-specific factors such as course difficulty, assessment methods, and students' prior

knowledge. Future research should examine how these behavioral patterns relate to performance outcomes and knowledge gains.

Finally, our study examined only two intervention time points. A more complete understanding of longitudinal seeding effects would require multiple interventions across longer time spans, potentially across multiple courses, to map how regulatory behaviors develop and stabilize over extended periods.

Despite these limitations, our findings demonstrate that self-regulated learning interventions create complex, temporally extended impacts that reshape the developmental trajectories of students' learning processes. By combining early skill-building with later corrective support, and by leveraging self-scheduling to create meaningful personal commitments, online courses can provide adaptive regulatory scaffolding that helps students develop the metacognitive capacities essential for success in self-directed learning environments. The challenge for future research and practice is to optimize these interventions in ways that enhance self-regulation while maintaining balanced engagement with the full range of learning activities that promote deep understanding.

6 Conclusions

This study demonstrates that self-regulated learning interventions in online courses operate through more complex and temporally extended mechanisms than typically recognized. Our findings reveal three key insights with important implications for both theory and practice.

First, we provide evidence for a **longitudinal seeding effect**: the initial survey-based intervention showed limited immediate behavioral changes but created conditions that altered students' developmental trajectories over weeks. By the time students reached the LoS phase, the experimental group had already adopted qualitatively different learning patterns, particularly in how they engaged with learning resources and assessment activities. This suggests that intervention research focusing exclusively on immediate post-intervention outcomes may substantially underestimate intervention impacts (Koedinger et al., 2013; Qun, 2025). Early interventions may work not primarily by producing immediate behavioral corrections but by establishing metacognitive frameworks and regulatory dispositions that shape how students' learning processes evolve over time.

Second, our LoS intervention triggered what we term a **metacognitive awakening**. The experimental group's significant revision of their deadline perceptions ($p < .001$), combined with 6.75-fold increases in feedback-seeking behaviors and ninefold increases in self-assessment activities, suggests the intervention prompted fundamental reassessment of learning processes rather than simply increased effort. Importantly, this occurred in a context where students had set their own schedules—being notified of lateness relative to one's own commitments appears to create stronger metacognitive reflection than missing externally imposed deadlines. This finding aligns with goal-setting theory but extends it by demonstrating that the personal ownership created by self-scheduling can be leveraged to enhance intervention effectiveness.

Third, our detailed behavioral analysis reveals important **trade-offs and side effects** that emerge from intervention. While students in the experimental group developed more sophisticated self-regulatory patterns with reflective activities serving as central hubs in their learning networks, they engaged less frequently with extended learning resource sequences. The control group's sustained focus on deep, repetitive content engagement represents a different but not necessarily inferior approach. This highlights that interventions do not simply add beneficial behaviors but fundamentally redistribute how students allocate limited time and attention across learning activities. The implications are nuanced: interventions successfully promoting self-regulation may inadvertently reduce other valuable learning behaviors.

These findings contribute to self-regulated learning theory by demonstrating that interventions create cascading effects across time and across behavioral systems rather than isolated, immediate impacts. For learning analytics research, our results illustrate the value of sequence and network analysis for revealing the structural complexity of learning processes—differences between groups were evident in behavioral patterns and network topology even when overall activity levels appeared similar.

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Appendix A

Example of a student's self-assessment scores and the automated feedback received. The low scores are marked with **bold and underline**.

Statement	Score (0-10)
I think about what I really need to learn before I begin a task in this online course. (Prethink)	**3**
I set standards for my assignments in online courses. (Standards)	**3**
I usually self-assess my performance once I finish. (SelfAssess)	**2**
I set short-term (daily or weekly) and long-term (monthly or semester) goals. (Goals)	**2**
I summarize my learning in online courses to examine my understanding of what I have learned. (Summary)	**3**
I am aware of what strategies I use when I study for this online course. (Strategies)	**1**
I plan out my schedule each week, so I have the appropriate amount of time available for online study. (Schedule)	**3**
I choose the location where I study to avoid too much distraction. (Location)	7
I choose a time with few distractions for studying for my online courses. (Time)	8
I am persistent in getting help from the instructor through e-mail. (Askhelp)	7
I set realistic deadlines for learning. (Deadlines)	6
I think about how I might improve my work by evaluating it against the marking criteria provided by the teacher. (Criteria)	7

The prompt given to the student based on the item scores below the mid-point (5) of the scale (0-10).

Hi (student's first name),

(Prethink) Before starting a study session, think about what you want to do. Get an overview of the instructions and study materials to get the big picture. **(Standards)** Think about what you want to achieve: are you aiming for perfect performance or is the minimum performance enough for you to complete the task/course?

(Self-Assess) Maybe you can look at a previously returned assignment submission. What does your performance look like? Can you adjust your doing based on previous feedback? If possible, compare your submission to the evaluation criteria. What areas for development do you notice? Have you achieved what you aimed for?
(Goals) Goal setting can be done in the short or long term. Think about what you want to achieve per study session, individual assignment, and course. Break larger goals into smaller, easier-to-perceive and manageable parts.
(Summary) At the end of the study session, recall what you wanted to learn. Review your learning objectives. Did you achieve your goal? Did you understand everything? If you are unsure about any area, study it again and, if necessary, look for more information on the topic.
(Strategies) Get an overview of the instructions and study materials to get the big picture. Break larger goals into smaller, easier-to-perceive and manageable parts. When you familiarize yourself with the learning materials, think about how they relate to things you previously knew. Can you come up with your own examples where you can apply the models or concepts presented? Make summaries describing the topic in your own words. Think critically: what does the new information mean and do you agree with the presenter?
(Schedule) Consider using a calendar for weekly scheduling. Take into account different areas of life (study, work, family, hobbies). Reserve time for studying every week. Check at the end of the week how much time you have spent studying and what you have achieved. Modify the schedule for the next week if necessary so that you can stick to the deadlines. Check your notes, assignment feedback, and prepared schedules. Did you manage to stick to your plan? If you recognize a point that went differently than planned, think about why that happened. Was there a reason you couldn't foresee? Can you somehow avoid a similar situation happening later?

Appendix B

The diary variables, original statements used to derive the variables, and the source of original statements.

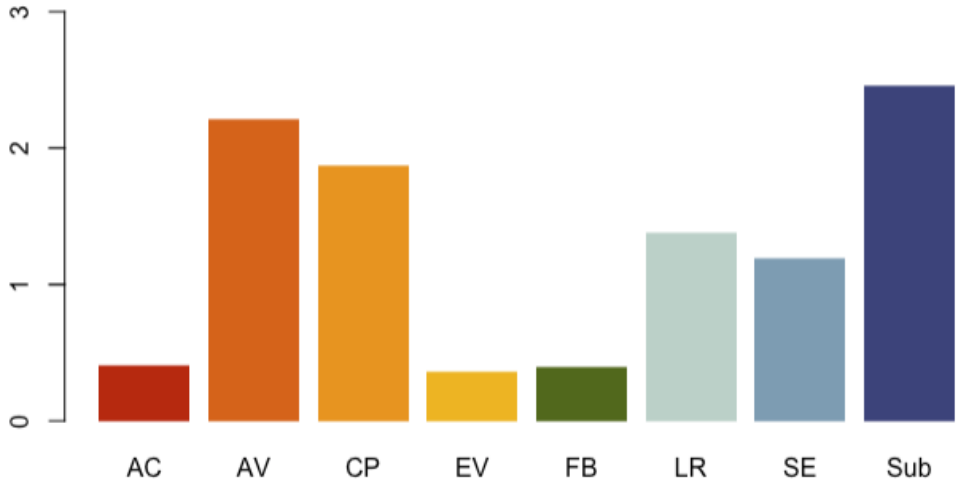
Variable	Statement	Source
Askhelp	“I am persistent in getting help from the instructor through e-mail.”	OSLQ (Barnard et al., 2009)
Criteria	“I think about how I might improve my work by evaluating it against the marking criteria provided by the teacher.“	SRL-O (Broadbent et al., 2023)
Deadlines	“I set realistic deadlines for learning.”	SRL-O (Broadbent et al., 2023)
Goals	“I set short-term (daily or weekly) and long-term (monthly or semester) goals.”	OSLQ (Barnard et al., 2009)
Prethink	“I think about what I really need to learn before I begin a task in this online course.“	SOL-Q (Jansen et al., 2017)
Schedule	“I plan out my schedule each week, so I have the appropriate amount of time available for online study. “	SRL-O (Broadbent et al., 2023)
SelfAssess	“I usually self-assess my performance once I finish.“	SRL-O (Broadbent et al., 2023)
Standards	“I set standards for my assignments in online courses.”	OSLQ (Barnard et al., 2009)
Strategies	“I am aware of what strategies I use when I study for this online course.”	SOL-Q (Jansen et al., 2017)
Summary	“I summarise my learning in online courses to examine my understanding of what I have learned.”	OSLQ (Barnard et al., 2009)
Time	“I choose a time with few distractions for studying for my online courses.”	OSLQ (Barnard et al., 2009)
Location	“I choose the location where I study to avoid too much distraction.”	OSLQ (Barnard et al., 2009)

Appendix C

Coding scheme mapping Moodle log activities to high-level learning states.

Learning State	Moodle Component	Event Name / Action	Description & Coding Principle
Assessment Criteria (AC)	Page	Course module viewed	Viewing specific pages dedicated to grading rubrics, assessment guides, or project requirements.
Assignment Viewed (AV)	Assignment	The user has viewed the status of the submission	Viewing the assignment landing page, instructions, or deadlines without submitting.
Course Page	System / Course	Course viewed	Viewing the main course landing page, dashboard, or course module list.
Evaluating	Assignment	Assignment graded	Teacher has evaluated the student's submission.
Feedback	Gradebook / Assignment	User report viewed, Grading table viewed	Viewing the gradebook, individual grades, or instructor feedback comments on submissions.
Learning Resource Viewed (LRV)	File, Folder, URL, Page	Course module viewed	Consumption of instructional materials, including opening PDF files, external links, or content pages.
Self-Evaluation (SE)	Feedback (Activity)	Feedback viewed, Response submitted	Interacting with the specific "Feedback" activity tool used for the self-reflection surveys (viewing questions or submitting answers).
Submitting	Assignment	A file has been uploaded, Submission created, Submission confirmed	Active steps involved in uploading a document or finalizing a submission for grading.

Appendix D



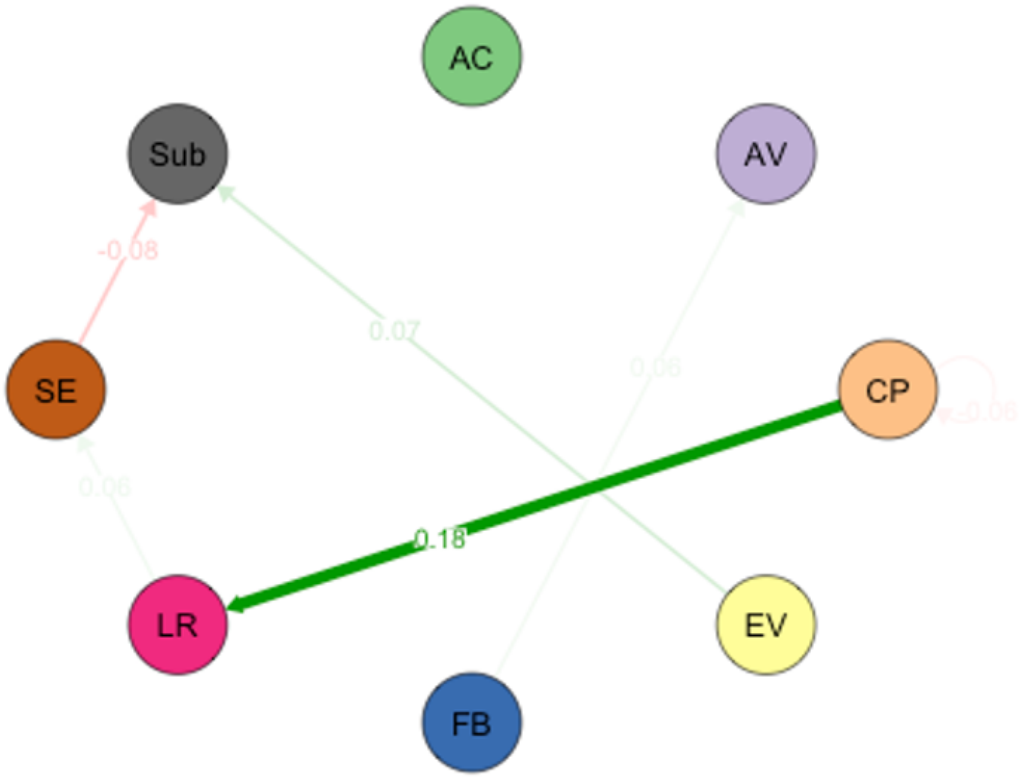
Mean time students spent engaged in each of the eight defined learning states (AssessmentCriteria (Red), Assignment viewed (Dark Orange), Course page (Light Orange), Evaluating (Yellow), Feedback (Green), Learning Resource viewed (Light Blue), Self-evaluation (Medium Blue), Submitting (Dark Blue)) across all 9,059 observed sequences. The y-axis represents an aggregated time measure.

Appendix E



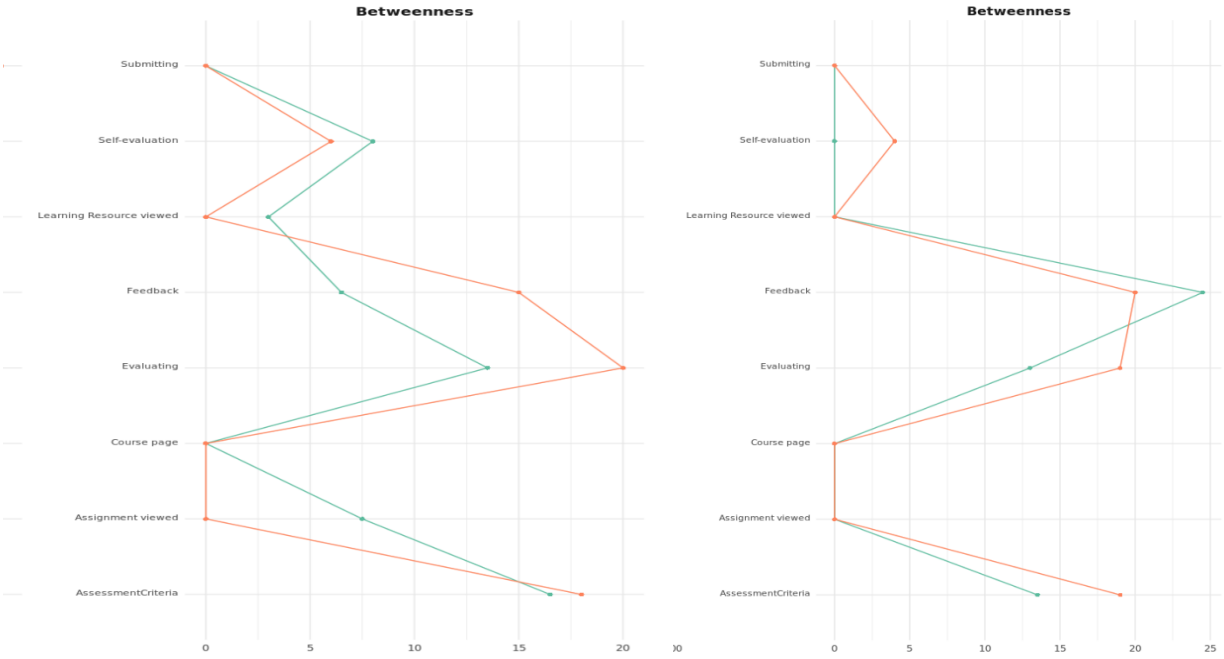
Bar charts showing the absolute frequency of each learning state before (green) and after (orange) the intervention for the Control group (left) and the Experimental group (right). The Experimental group (right) shows a disproportionate growth in "Self-evaluation" and "Feedback" compared to the Control group (left), whose growth is more evenly distributed across "Assignment viewed" and "Submitting." This visualizes the "strategic recalibration" discussed earlier, where the Experimental group shifted effort toward regulatory activities.

Appendix F



Permutation test results showing changes within the Control group (mostly red) and Experimental group (mostly green) following the LoS intervention period.

Appendix G



Parallel coordinate plots comparing betweenness (a state's importance as a bridge in the network) measures for each learning state. The plots show comparisons between Control (green) vs. Experimental (orange) before (left) and after (right) the intervention.